

Foodz@Yourz Database Design & Data Modelling

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1. ENTITIES

Entity Name	Brief Explanation
Product	Represents the product offered by Foodz@Yourz. Each product is offered by one or more store
Store	Represents a location/outlet where products are sold or stored.
Collection	Represents a set of possibly deliveries.
Perishable	A subclass of Product that has an expiration date.
Non_Perishable	A subclass of Product that doesn't have an expiration date.
Food	A subclass of Perishable product
Drink	A subclass of Perishable product
Soft_Drink	A subclass of Non Perishable product
Alcoholic_Bev	A subclass of Non Perishable product
Member	Represents a registered customer.
Basket	Represents a shopping cart created by a member.
Payment	Represents a payment transaction for a basket.
Card_Pay	A subclass of payment using a card
Mobile_Pay	A subclass of payment using mobile platform
Small_Coll	A subclass of Collection with a maximum order limit.
Large_Coll	A subclass of Collection with a minimum order limit.
Order	Represents the order confirmed by customer
Delivery	Represents the delivery details of the customer who placed the order
Address	Represents the home address of the customer
Employee	Represents a person employed by Foodz@Yourz.
Driver	A subclass of employee who drives vehicle
Warehouse_Worker	A subclass of Employee who works in the warehouse/store.
Vehicle	Represents the vehicle used for deliveries

Two_W	Subclass of vehicle used to deliver the small collection of orders
Four_W	Subclass of vehicle used to deliver the large collection of orders

2. SPECIALISATIONS

General Entity	Specialised entity	Brief explanation
Product	Perishable, Non Perishable	Products are grouped according to whether or not they expire. "Best before" and "use by" dates are necessary for perishable goods but not for non-perishable ones.
Perishable	Food, Drink	Food with dietary and allergen tags and drink with serving volume attributes are two more categories for perishable goods.
Non Perishable	Soft_Drink, Alcoholic_bev	Soft drinks containing sugar and alcoholic beverages including alcohol % and legal age limits are two categories for non-perishable goods.
Collection	Small_Coll, Large_Coll	Size restrictions are used to categorise collections. There is a minimum restriction for large collections and a maximum limit for small collections.
Payment	Card_Pay, Mobile_Pay	Payments are categorised according to the settlement method. Whereas mobile phone

		Pay records the mobile phone number, Card Pay records the card data.
Vehicle	Two_W, Four_W	vehicles are classed according to the number of wheels they have.
Employee	Driver, Warehouse_Worker	Workers are identified by their main function. While warehouse employees have a set shift, drivers use vehicles and have a licence number.

3. RELATIONSHIPS & MULTIPLICITIES

Entity Name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justifications for the multiplicity
Member	1..*	creates	0..1	Basket	1. Each Member may create one or more Baskets. 2. Each Basket must be created by exactly one Member. 3. A Basket cannot exist without being linked to a Member, showing mandatory participation on the Basket side. 4. The Member side is optional because a member may register but not yet create a basket.

Basket	1..1	based on	1..1	Order	1. Each Order must be based on one Basket. 2. Each Basket can result in only one Order. 3. The relationship is mandatory for both entities, as an Order cannot exist without a Basket. 4. This ensures referential integrity between shopping and ordering processes.
Order	1..1	can have	1..1	Delivery	1. Each Order must have one corresponding Delivery. 2. Each Delivery relates to exactly one Order. 3. This relationship is mandatory for both entities, as every order must be delivered. 4. The one-to-one relationship ensures tracking of delivery per order.
Store	1..*	has	1..*	Product	1. Each Store may hold multiple Products. 2. Each Product may be stocked in more than one Store. 3. The relationship is many-to-many, reflecting inventory distribution across branches. 4. Both sides have mandatory participation to ensure product availability.
Delivery	1..1	used for	0..*	Address	1. Each Delivery must be associated with exactly one Address (1..1). 2. An Address may be used for zero or many Deliveries (0..*), e.g., repeat customers or unused saved addresses.

					3. Addresses are shared resources across deliveries, so the address side is many/optional. 4. The mandatory side is Delivery, because a delivery must have a destination
Order	1..1	settled by	1..1	Payment	1. Each Order is settled by exactly one Payment transaction (1..1). 2. Each Payment record belongs to exactly one Order (1..1). 3. The diagram shows this as mandatory on both sides to ensure a completed payment is recorded for every order. 4. This provides accountability and an auditable payment per order.
Payment	1..1	specialisation of	{Mandatory, Or}	Card_Pay / Mobile_Pay	1. Payment must be one of the specialised methods (Card or Mobile) {Mandatory, Or} as in the diagram. 2. Exactly one specific payment subtype (Card_Pay or Mobile_Pay) records method-specific attributes. 3. The mandatory or constraint ensures a payment record cannot exist without specifying its type. 4. This models alternative method types with required choice of one subtype.
Product	{Mandatory, Or}	specialised into	{Mandatory, Or}	Perisable / Non_Perisable	1. Product is specialised into Perisable or Non_Perisable 2. The diagram shows a mandatory choice (product must be classified) {Mandatory, Or}

					3. Within Perisable, further subtypes (Food/Drink) exist, and some specialisations are {Mandatory, And} where attributes from both may apply (diagram markings). 4. This enforces correct product-type attributes (dates, use_by etc.) only when applicable.
Drink	{Mandatory, And}	specialised into	1..*	Soft_Drink / Alcoholic_bev	1. Drink may be further specialised into Soft_Drink and Alcoholic_bev (diagram shows {Mandatory, And}, meaning drink-type attributes may require additional subtype attributes). 2. Alcoholic_bev includes percentage and legal_min_age attributes. 3. Soft_Drink includes sugar_100ml attribute. 4. The specialisation enforces the presence of correct extra attributes depending on the subtype.
Collection	1..*	can have	1..1	Order	1. Each Collection (Small or Large) can include one or more Orders (1..*). 2. Each Order must belong to exactly one Collection (1..1), ensuring grouping for delivery or promotion. 3. The relationship is mandatory for Order, as shown by the solid line (each order must be part of a collection). 4. The Collection side allows multiple orders, enabling batch processing or grouped dispatches.
Two_W	1..1	can have	1..*	Small_Coll	1. Each Two_W vehicle can handle multiple Small Collections (1..*).

					2. Each Small_Collection is assigned to exactly one Two_W vehicle (1..1). 3. This represents capacity and practicality — small vehicles can deliver smaller bundles. 4. The relationship is mandatory for Small_Collection, as shown by the solid connector in the EERD.
Four_W	1..1	assigned	1..*	Large_Coll	1. Each Four_W (four-wheeler) can assign deliveries to multiple Large_Collections (1..*). 2. Each Large_Collection is assigned to exactly one Four_W vehicle (1..1). 3. The relationship represents large delivery operations needing higher capacity vehicles. 4. It's mandatory for Large_Collection, ensuring proper vehicle assignment for transportation.
Driver	1..1	uses	1..*	Vehicle	1. Each Driver may use one or more Vehicles across different shifts (1..*). 2. Each Vehicle is used by exactly one Driver at a time (1..1). 3. This ensures operational accountability for each delivery run. 4. The relationship is mandatory for Vehicle, since no vehicle can operate

					without a driver.
Basket	1..*	contains	1..*	Product	1. Each Basket can contain multiple Products (1..). 2. Each Product can appear in multiple Baskets (1..). 3. This many-to-many relationship captures the customer's selection of several products per basket. 4. It is mandatory for both entities to represent valid basket product combinations in the ordering process.

4. ATTRIBUTES & PRIMARY KEYS

Entity Name	Attributes for each entity (include PK)	Brief explanation
Product	p_id (PK), p_name, p_category, p_sku	The product's unique identifier (PK), name, description, and stock-keeping unit are some of its most important characteristics.
Perishable	best_before, use_by	Unique characteristics for perishable goods, including required "best before" dates and flexible "use by" dates.
Non_Perishable	(No any unique attributes)	Represents the Non Perishable food category Inherits all attributes from Product
Food	diet_tags, allergens	Details about food things, such as allergen warnings and dietary restrictions (tags).
Drink	ml_per_serv	A specific property for drinks that measures the amount of liquid in a serving in millilitres.
Soft_Drink	suger_100ml	This particular soft drink attribute describes the amount of sugar in 100 milliliters.
Alcoholic_beve	percentage, legal_min_age	describing the minimum legal buying age and the proportion of alcohol

Store	s_id (PK), s_name, s_address, s_type	characteristics that define the store or outlet, such as its name, physical address, kind (e.g., small, large), and unique identification (PK).
Member —	m_id (PK), m_name, m_email, regis_at	Features for consumers who have registered, such as their name, login timestamp, email address, and unique identification (PK).
Basket	b_id (PK), b_name, creation_date, b_status	characteristics that keep track of the shopping basket, such as its name, creation date, unique identifier (PK), and current status (e.g., active, finalised).
Payment	pay_id (PK), pay_type, pay_date, pay_time	features that track payment transactions, such as the date, type (credit, mobile, etc.), unique identification (PK), and transaction time.
Card_Pay	last_four_digit, card_holdername	particular characteristics for card payments, including the last four digits and the cardholder's name for security.
Mobile_Pay	mobile_num	Recording the mobile phone number used for the transaction is a unique feature for mobile payments.

Order	o_id (PK), o_date, o_status, o_type	characteristics of a verified customer order, such as its date, status, type (e.g., standard, express), and unique identifier (PK).
Delivery	delv_id (PK), delv_status, delv_type	Attributes tracking the shipment, including its unique identifier (PK), current status, and type (e.g., standard, scheduled).
Address	addr_id (PK), street, province, country	characteristics that define a physical location, such as its country, province, street details, and unique identification (PK).
Employee	e_id (PK), e_name, e_jobtitle, e_contact, e_address	characteristics that define an employee, such as their name, work title, contact details, primary address, and unique identification (PK).
Driver	d_license_no, d_license_type	particular characteristics for workers in the driver position, including their licence type and unique licence number.
Warehouse_Worker	shift	particular characteristic for warehouse workers that describes their designated work shift.
Vehicle	v_id (PK), v_regist_num, v_capacity	characteristics that define a vehicle, such as its capacity (such as volume or weight), registration

		number, and unique identification (PK).
Two_W	(No unique attributes)	represents the category of two-wheeled vehicles. inherits every characteristic of the vehicle.
Four_W	(No unique attributes)	represents the category of four-wheeled vehicles. inherits every characteristic of the vehicle.
Collection	c_id (PK), c_date, c_time, c_status	characteristics that keep track of the collection or fulfilment event, such as its date, time, and current state, as well as its unique identifier (PK).
Small_Coll	max_order_limit	particular feature that specifies the highest value or size permitted per order for tiny collections.
Large_Coll	min_order_lim	particular feature that specifies the minimum size or value needed for each order in huge collections.

5. QUALITY & NOTATIONS OF CONCEPTUAL EERD

